

BARSTOW-DAGGETT, CA, USA

WMO: 723815

Lat: 34.854N

Lon: 116.786W

Elev: 584

StdP: 94.50

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23161

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.3	0.3	-18.8	0.8	9.9	-17.0	0.9	9.3	15.5	12.6	14.3	12.8	2.5	270	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	42.7	20.0	41.4	19.4	40.2	18.8	22.5	36.9	21.6	36.2	20.8	35.5	4.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	14.2	27.6	16.8	12.8	29.3	14.7	11.2	31.4	68.7	37.1	65.5	35.9	62.5	35.7	33.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.4	11.1	-4.8	45.2	2.1	1.1	-6.3	46.0	-7.5	46.6	-8.7	47.3	-10.2	48.1		
(5)			-7.3	24.6	1.9	2.8	-8.6	26.6	-9.7	28.2	-10.7	29.8	-12.1	31.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.3	9.8	11.6	15.5	18.7	23.5	29.0	32.2	31.6	27.4	20.5	13.8	9.0
	DBStd	8.86	3.25	3.59	3.88	4.26	4.37	3.75	2.36	2.25	3.46	3.93	4.02	3.19
	HDD10.0	139	43	22	3	1	0	0	0	0	0	0	12	58
	HDD18.3	1069	264	188	104	47	9	1	0	0	1	23	142	289
	CDD10.0	3884	37	68	174	261	420	569	689	668	522	324	126	27
	CDD18.3	1772	0	1	16	58	170	320	431	410	272	89	6	0
	CDH23.3	25822	2	28	269	764	2231	4719	6778	6251	3616	1049	112	2
CDH26.7	16016	0	2	65	308	1208	3047	4601	4162	2155	447	21	0	
Wind	WSAvg	4.8	3.5	4.3	5.2	6.3	6.5	6.2	5.1	4.8	4.5	4.2	3.7	3.4
Precipitation	PrecAvg	100	12	15	12	5	2	3	9	8	12	4	7	9
	PrecMax	190	63	64	48	46	12	32	52	34	85	26	44	51
	PrecMin	31	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	41	15	18	13	10	3	6	11	10	21	6	10	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.6	26.2	31.3	35.0	39.9	43.8	44.8	43.5	40.4	35.7	28.8	22.6
		MCWB	11.3	11.6	13.9	15.7	18.6	18.5	20.8	21.0	19.0	16.8	12.9	11.7
	2%	DB	20.6	23.9	28.4	32.5	37.4	41.6	43.1	42.2	38.9	33.2	26.4	19.7
		MCWB	9.8	10.9	12.8	14.5	17.2	18.5	20.3	19.9	18.6	15.8	12.0	9.9
	5%	DB	18.8	21.6	26.3	30.3	35.4	40.1	41.9	40.9	37.5	31.3	24.1	17.7
		MCWB	8.9	10.1	11.9	13.5	15.9	18.0	19.8	19.2	18.3	14.8	11.4	8.8
	10%	DB	16.9	19.2	24.2	27.9	33.2	38.5	40.4	39.5	36.1	29.1	21.8	15.7
		MCWB	8.2	9.1	11.3	12.7	15.3	17.4	19.3	18.7	17.4	13.8	10.4	7.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.6	14.9	16.8	20.7	22.4	24.0	23.5	21.9	18.8	14.6	13.2
		MCDB	19.8	22.0	26.7	32.4	38.2	40.5	39.0	37.8	33.7	29.4	24.3	19.8
	2%	WB	11.5	12.1	13.6	15.3	18.0	20.3	22.7	22.3	21.0	17.1	13.2	11.6
		MCDB	17.2	20.5	25.5	30.3	34.7	38.9	37.1	36.8	33.3	29.6	23.7	17.4
	5%	WB	10.3	11.0	12.7	14.1	16.6	19.0	21.8	21.4	20.2	15.8	12.0	10.2
		MCDB	15.9	19.1	24.4	28.1	33.1	37.6	36.5	36.1	32.8	28.8	22.4	15.8
	10%	WB	9.2	10.0	11.8	13.1	15.6	17.9	20.9	20.5	19.1	14.8	11.1	8.8
		MCDB	15.1	17.7	22.8	26.3	31.5	36.3	35.9	35.5	32.3	27.1	20.5	14.0
Mean Daily Temperature Range	5% DB	MDBR	13.0	13.3	14.4	14.8	15.5	16.7	16.2	16.2	15.8	15.1	14.2	12.7
		MCDBR	15.4	16.5	17.0	17.5	17.6	18.0	17.5	16.9	16.6	17.0	16.4	14.9
		MCWBR	7.9	7.6	6.9	6.8	6.3	6.1	6.2	6.1	6.1	7.1	7.8	8.1
	5% WB	MCDBR	12.3	14.0	15.7	16.3	16.9	17.5	15.1	15.0	14.8	15.4	15.0	12.6
		MCWBR	6.7	6.8	6.6	6.5	6.8	7.0	6.8	6.7	5.9	6.5	7.2	7.0
Clear-Sky Solar Irradiance	taub	0.273	0.282	0.294	0.326	0.336	0.337	0.371	0.351	0.330	0.309	0.275	0.274	
	taud	2.546	2.534	2.492	2.391	2.391	2.405	2.376	2.417	2.456	2.489	2.576	2.565	
	Ebn at Noon	948	974	984	959	946	941	906	922	930	926	939	926	
	Edh at Noon	79	89	101	118	120	118	121	114	104	92	76	73	
All-Sky Solar Radiation	RadAvg	3.24	4.23	5.72	7.08	7.95	8.51	7.76	7.27	6.33	4.87	3.63	2.93	
	RadStd	0.23	0.36	0.23	0.22	0.29	0.30	0.35	0.26	0.20	0.23	0.14	0.18	

Historical Trends

		Heating		Cooling		Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (40 neighbors)		N/A	N/A	-1.67	N/A	-0.54	N/A	N/A	N/A	N/A
		+0.62	N/A	-1.83	+0.57	N/A	N/A	-99	+217	+117

Nomenclature: See separate page